

Exercise 7: Multivariate Optimization 2

Optimization in Machine Learning

Reproducibility

Produced with Python 3.10 and R 4.4.1. To reproduce, install the pinned package versions below.

Python

```
pip install numpy==2.0.2 matplotlib==3.10.8
```

R

```
install.packages("remotes")  
remotes::install_version("ggplot2", "3.5.1")
```

Setup

A radial basis function (RBF) network has been fitted to a black-box function, giving the model

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, \mathbf{x} \mapsto \sum_{i=1}^2 w_i \cdot \rho(\|\mathbf{x} - \mathbf{c}_i\|_{S_i})$$

with $\mathbf{c}_1 = (-1.1, 1.1)^\top$, $\mathbf{c}_2 = (0.8, -0.8)^\top$, weights $w_1 = 1$, $w_2 = -1$, **biweight (quartic) kernel**

$$\rho : \mathbb{R} \rightarrow \mathbb{R}, u \mapsto \begin{cases} (1 - u^2)^2 & |u| < 1 \\ 0 & \text{otherwise} \end{cases}$$

and Mahalanobis distances $\|\cdot\|_{S_i}$ with covariance matrices $S_1 = \mathbf{I}$, $S_2 = \begin{pmatrix} 1.1 & -0.9 \\ -0.9 & 1.1 \end{pmatrix}$. The Mahalanobis distance is $\|\mathbf{x} - \mathbf{c}\|_S = \sqrt{(\mathbf{x} - \mathbf{c})^\top S^{-1}(\mathbf{x} - \mathbf{c})}$.

Note: We chose this kernel and distance for educational purposes; in practice, Gaussian kernels and Euclidean distances are more common.

(a) Surface plot

Plot f on the range $[-2, 2] \times [-2, 2]$.

(b) Disjoint supports

Show that $\bigcap_{i=1}^2 \{\mathbf{x} \in \mathbb{R}^2 \mid \rho(\|\mathbf{x} - \mathbf{c}_i\|_{S_i}) \neq 0\} = \emptyset$.

(c) Global minimum, analytically

Find the global minimum of f analytically.

Hint: recall (b).

(d) Two GD steps numerically

Write an R/Python script that performs two gradient descent steps starting at $\mathbf{x}^{[0]} = (-0.45, 0.5)^\top$ with step size $\alpha = 0.15$. What do you observe?

(e) Two GD steps analytically

Confirm (d) with paper-and-pencil arithmetic. Same $\mathbf{x}^{[0]} = (-0.45, 0.5)^\top$, $\alpha = 0.15$.

(f) Gradient descent with momentum

Repeat (d) but with **gradient descent with momentum**. Use velocity initialization $\mathbf{v}^{[0]} = (0.4, -0.4)^\top$, momentum coefficient $\varphi = 0.5$, and run for 15 iterations with step size $\alpha = 0.15$. Find the global minimum.